

vision**2**learn

Coding Practices Level 3

Hints
and
Tips

Unit 5



Contents

Section 1 – Know about deployment	4
Section 2 – Understand maintenance in coding	6
Section 3 – Understand configuration management and version control systems	7

Hints and Tips Unit 5

The information in this guide will be helpful to you when you are ready to complete your assessment:

Unit 5: Understand deployment, maintenance and configuration management

Section 1 – Know about deployment

Q1. Explain what packages and artifacts are, and their relationship.

Within your explanation of what packages and artefacts are, you will also need to explain the relationship between them. You should also demonstrate an understanding of the process for how a package is created and its purpose within deployment.

Q2. Explain the importance of deployment planning.

As an explain question, you will need to provide detailed information for what deployment planning is and how it's used. You could include examples and also consider the advantages and disadvantages of deployment planning.

Q3. Explain the difference between build and deploy.

Within your explanation, think about when build and deploy are used, when and why would each be used and what's involved with each.

Q4. Describe what is meant by continuous deployment.

Within your description think about why and when it's used within the DevOps lifecycle, think about the differences between continuous delivery and continuous deployment. What are the advantages and disadvantages with this process?

Q5. Describe what is meant by deployment strategy.

Give some detail to your description by including some of the aims of deployment strategy. Consider if there's a time or place within the software product that this fits into best. Why should it be used?

Q6. Identify the following:

- a) the advantages and disadvantages of canary deployment
- b) what is meant by rollback strategy

a) Your answer must be more than a list of advantages and disadvantages. It's important that you demonstrate understanding of the term 'canary deployment' and provide a little detail for the advantages and disadvantages this form of deployment may have.

b) Although this an identity question, it still requires some detail to demonstrate your understanding of what rollback strategy is and you should therefore include what's included within it.

Section 2 – Understand maintenance in coding

Q1. Identify the four key elements of maintenance.

As an identify question you should add a little detail to each element that you list to demonstrate a basic understanding of each one.

Q2. Explain what types of documentation are important to Maintenance Engineers, including what should be created by Maintenance Engineers. Within your answer also consider:

- What is meant by 'live' documentation
- What the potential benefits are for keeping documentation
- What the potential pitfalls are for not keeping documentation

The bullet points in this question are there to help you include the detailed explanation that's needed to meet this criteria and you should therefore consider each carefully. Use examples where you can and be sure to include different types of documentation.

Q3. Explain the following:

- Improvements that could be made by a Maintenance Engineer
- Under what conditions they should be made
- Any potential consequences of carrying out the improvement work

As with question 2, ensure that you include each bullet point within your answer and provide a full explanation for each one. Again, examples will help with your answer.

Q4. Explain what change management processes might be followed whilst maintaining software and what benefits or potential problems could arise from the use of change management.

It's important to include all of the parts of this question within your answer so be sure that you read it carefully. You must include the processes for maintaining software, the benefits and disadvantages that can occur at any stage.

Section 3 - Understand configuration management and version control systems

Q1. Explain what is meant by configuration management

Within your explanation consider what's included in the process of configuration management, why it's used and when.

Q2. Explain the tools and techniques used in configuration management systems, including the benefits of SCM (Software Configuration Management)

Ensure that you include different tools and techniques within your answer. It may also be beneficial to explain when they might be used. When considering the benefits, you might also find it useful to explain some of the disadvantages and why one tool or technique might be better than another.

Q3. Explain the main features and benefits of version control systems. Include your understanding of what issues could arise for a Software Development team of not using version control processes.

Include why and how different version control systems are used and at what stage. What are the main features of each version control? Think about the difficulties that could arise if version control isn't used and how the team would be able to work together without it.